

High Availability — Guided Notes ANSWER KEY

Unit 3.2.1 | CompTIA Network+ N10-009 Obj. 3.2 | N10-008 3.2

For instructor use / distribute after students complete the notes.

Question 1

High availability is measured as a percentage of uptime. 'Five nines' (99.999%) allows approximately _____ minutes of downtime per year. Why do small improvements in availability percentage represent disproportionately large reductions in downtime?

Answer: 5.26 minutes. Because the percentage scale compresses at the high end — going from 99.9% to 99.99% cuts allowed downtime from ~8.7 hours to ~52 minutes, a 10x reduction for a 0.09% improvement.

Question 2

HSRP (Hot Standby Router Protocol) and VRRP (Virtual Router Redundancy Protocol) both provide _____ router redundancy by presenting a shared _____ IP and MAC address to clients. What happens to client traffic if the active router fails?

Answer: Default gateway; virtual. The standby router detects the failure and takes over the virtual IP and MAC within seconds — clients continue sending traffic to the same gateway address without reconfiguration.

Question 3

In a redundant network design, a _____ point of failure is a single device or link whose failure causes a complete outage. Name two common single points of failure in a school network and how redundancy eliminates each.

Answer: Single; examples: single uplink from access switch to core (fix: dual uplinks with EtherChannel or STP failover), single ISP connection (fix: dual ISP with BGP or failover routing), single firewall (fix: HA firewall pair).

Question 4

EtherChannel (LACP) bundles multiple physical links into one logical link, increasing _____ and providing _____. If one physical link in a four-link bundle fails, what percentage of bandwidth is retained?

Answer: Bandwidth (throughput); redundancy (link failure tolerance). 75% — three of four links remain active.

Question 5

A UPS (Uninterruptible Power Supply) provides battery power during an outage long enough for a _____ shutdown or _____ activation. Without a UPS, what specific type of data corruption risk exists when a switch or server loses power mid-operation?

Answer: Graceful; generator. File system corruption — writes that were in progress are incomplete, leaving the OS or application in an inconsistent state that may require manual repair or restore from

backup.

Question 6

Recovery Time Objective (RTO) defines the maximum acceptable _____ after a failure. Recovery Point Objective (RPO) defines the maximum acceptable _____ of data. A school that backs up data nightly has an RPO of approximately _____ hours.

Answer: Downtime (time to restore service); data loss (how old the most recent backup can be). 24 hours — if a failure occurs just before the next backup, up to one day of data could be lost.

Question 7

Spanning Tree Protocol (STP) prevents _____ in redundant switched networks by blocking one or more redundant ports. The tradeoff with STP redundancy is that blocked ports require _____ to transition to forwarding if the active path fails.

Answer: Broadcast storms (switching loops); time (STP convergence — up to 30-50 seconds with classic STP, faster with RSTP).

Question 8 — Real World Application

REAL WORLD: A school's entire network goes down when a single uplink cable is accidentally disconnected by a custodian. The outage lasts 45 minutes while staff locate a replacement cable. Identify the design flaw, name the redundancy solution, and describe what the failure experience would look like with that solution in place.

Answer: Design flaw: single uplink from the access/distribution switch to the core — a single point of failure. Solution: dual uplinks with EtherChannel or STP failover. With redundancy: the second uplink carries all traffic within milliseconds of the first link going down (EtherChannel) or within seconds (RSTP failover). Users experience a brief interruption or nothing at all rather than a 45-minute outage.